Section A: Multiple Choice Questions

ANSWERS: Rates and Equilibrium PAST SAC Questions

Items 1 and 2 refer to the following:

The following gaseous equilibrium is established at high temperatures in the presence of a finely divided nickel (Ni) catalyst

$$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g);\ \Delta H = +206\ KJ\ mol^{-1}$$

Question 1

A particular reaction is carried out using equal amounts of $CH_{4(g)}$ and $H_2O_{(g)}$. Which one of the following set of changes in conditions would lead to the greatest **increase** in the proportion of reactants converted to products

	Volume of reaction vessel	Temperature
A.	increased	increased
B.	increased	decreased
C.	decreased	increased
D.	decreased	decreased

Question 2

This reaction occurs at a measureable rate only when the finely divided catalyst is present. This catalyst increased the reaction rate because

A. it weakly attracts the reaction products, driving the reaction to the right

B. the reactants can become attached to its surface where they can meet and undergo reaction.

C. it provides energy to the reactants when their molecules bounce off it, increasing the proportion of molecules in the gas state with the required activation energy.

D. It increased the equilibrium constant of the reaction, causing an increase in the proportion of products at equilibrium .

Question 3

Four different flasks A, B, C and D, at the same temperature, contain a mixture of PCl_5 PCl_3 and Cl_2. The concentration, in mol L^{-1}, of these components in each of the flasks is shown below.
In three of the four flasks, the mixture of gases is at equilibrium.
In which one is the mixture of gases **not** at equilibrium?

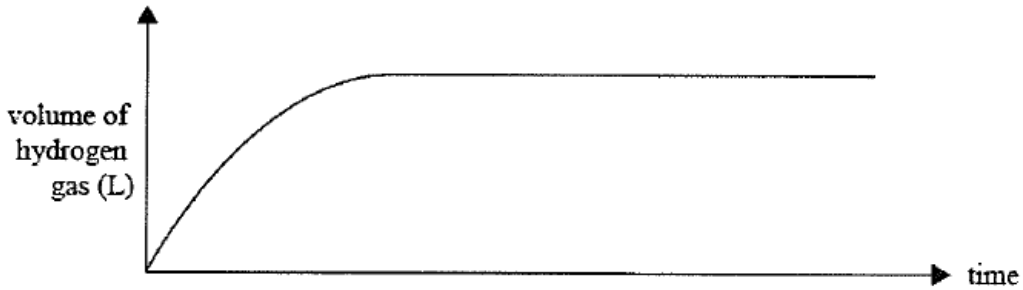
The equation for the reaction occurring is

$$PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$$

Flask	$[PCl_5(g)]$ M	$[PCl_3(g)]$ M	$[Cl_2(g)]$ M
A.	0.15	0.20	0.30
B.	0.20	0.15	0.15
C.	0.10	0.10	0.40
D.	0.30	0.80	0.15

Question 4

A 2.0 g piece of magnesium ribbon was added to a known volume of **2.0 M** hydrochloric acid. The volume of hydrogen gas produced during the reaction was measured and recorded.
The graph below shows the result of this experiment.



In another experiment 2.0 g of magnesium ribbon was added to the same volume of **1.0 M** hydrochloric acid, the graph of the profile for the production of hydrogen gas would *(in both cases the HCl was in excess)*

A. Be the same as in the first experiment
B. Have a steeper gradient at the start and plateau at the same level.
C. Have the same gradient at the start but plateau at a lower level.
D. Have a less steep gradient and plateau at the same level.

Question 5

For the reaction

$$A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)},$$

0.50 mole of A and 0.40 mole of B are placed into a 2.00 L closed reaction vessel at 25°C. After equilibrium has been established it was found that the container held 0.10 mol of product C.

The equilibrium constant for this reaction at 25° C is

A. 0.13 M^{-1}
B. 5.38 M^{-1}
C. 1.67 M^{-1}
D. 12.2 M^{-1}

Question 6

When a catalyst is added to a system at equilibrium, which of the following will decrease?

A. Heat of reaction
B. Activation energy
C. Potential energy of the reactants
D. Potential energy of the products

Question 7

For the reaction,

$$2HF(g) \rightleftharpoons F_2(g) + H_2(g) \qquad \Delta H = +536 \text{ kJ / mol}$$

Which of the following would cause a shift to the right?

A. adding F_2
B. removing HF
C. increasing pressure
D. increasing temperature

Question 8

The following equilibrium is established in a 2.0 L vessel:

$$2N_2(g) + O_2(g) \rightleftharpoons 2N_2O(g)$$

Which one of the following sets of conditions would be expected to push the equilibrium furthest to the right?

	Addition of chemical	**Volume of reaction vessel**
A.	nitrogen gas	increased
B.	nitrogen gas	decreased
C.	argon gas	increased
D.	argon gas	decreased

Question 9

A number of factors will increase the rate of a reaction.

I adding a catalyst
II increased temperature
III increased concentration of reactants
IV increased surface area of reactants

The factors that cause an increase in the reaction rate **solely** as the result of an **increase in the frequency of collisions** between particles are

A. III and IV only
B. I and IV only
C. II, III and IV only
D. I, II, III and IV

Question 10

For the reaction $2N_2(g) + O_2(g) \rightleftharpoons 2N_2O(g)$

Three different flasks, X, Y and Z contain a mixture of N_2, O_2 and N_2O at equilibrium. The concentrations, in molar, of these components in each flask is shown below

Flask	$[N_2(g)]$	$[O_2(g)]$	$[N_2O(g)]$
X	0.30	0.40	0.15
Y	0.60	0.10	0.15
Z	0.20	0.35	0.15

Which of the flasks is at a **different temperature** compared to the other two?

A. X
B. Y
C. Z
D. Unable to be determined from the information given.

Question 11

A mixture of ethane gas and hydrogen gas was allowed to react in a sealed 500 mL flask. The equilibrium that resulted may be represented as

$$C_2H_4\,(g) + H_2(g) \rightleftharpoons C_2H_6\,(g)$$

At equilibrium, there was found to be 1.25×10^{-1} mol of C_2H_6 (g), 7.50×10^{-2} mol C_2H_4 (g) and 8.35×10^{-2} mol of H_2(g) present in the flask.

The equilibrium constant for this reaction under these conditions is

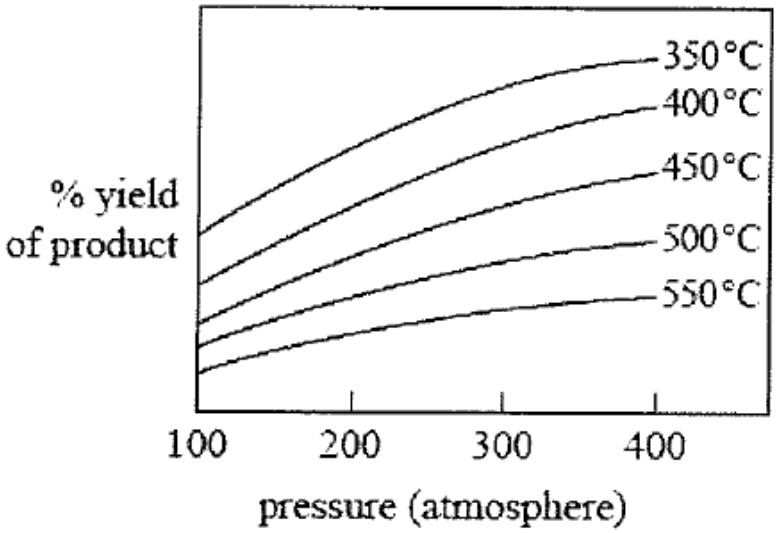
A. 0.0501
B. 0.100
C. 9.98
D. 20.0

Question 12

The graph below refers to the following gaseous reaction;

$$aA(g) + bB(g) \rightleftharpoons cC(g) + dD(g)$$

The effect of increasing pressure and temperature on the equilibrium yield of the products is shown in the graph below.



Which one of the following statements about the relative number of reactant and product molecules in the balanced equation is true?

A. The number of reactant molecules is equal to the number of product molecules.

B. The number of reactant molecules is greater than the number of product molecules.

C. The number of reactant molecules is less than the number of product molecules.

D. The relative number of reactant and product molecules cannot be determined from the data provided.

Question 13

The following reaction systems are at equilibrium in separate sealed containers.
The **volumes** of the containers are halved at constant temperature.
Which reaction has the largest percentage change in the concentration fraction (reaction quotient) immediately after the volume change?

A. $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

B. $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

C. $2CO_2(g) \rightleftharpoons 2CO(g) + O_2(g)$

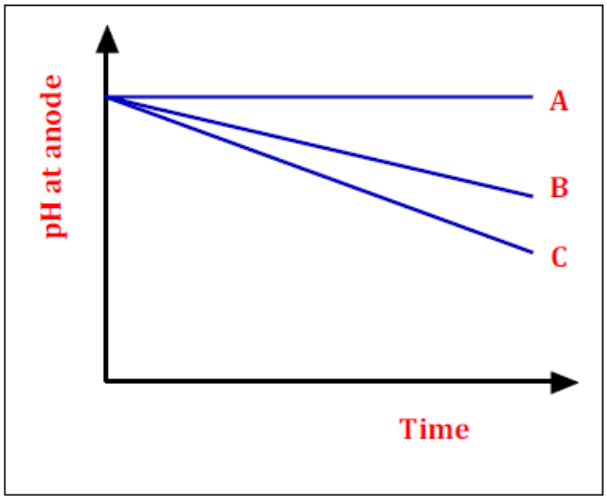
D. $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$

ANSWERS: Electrolysis PAST SAC Questions

Question 14

An electrolytic cell was constructed using different concentrations of sodium chloride and Platinum Electrodes.

The changes in pH that occur **at the anode** may be represented by lines A, B and C in the graph:

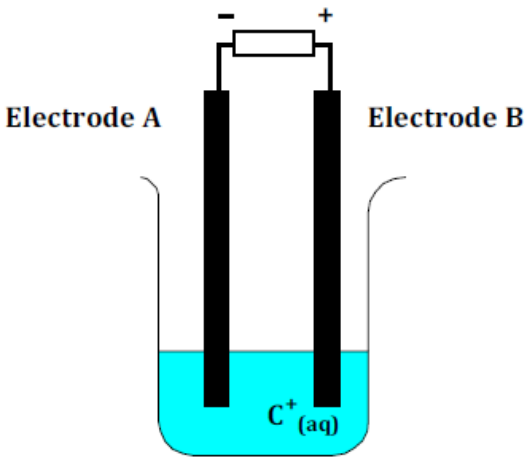


Which option below correctly represents the change in pH that occur **at the anode** at the different concentrations of *NaCl* ?

	1.0 M NaCl	**2.0 M NaCl**	**10.0 M NaCl**
a)	Line A	Line B	Line C
b)	Line C	Line B	Line A
c)	Line A	Line C	Line B
d)	Line C	Line A	Line B

Question 15

Consider the following ***Electrolytic Cell*** on the right:



Which of the following statements is correct?

a) Metal C would be deposited at electrode A.

b) Electrons flow through the electrolyte from A to B.

c) Electrode A represents the anode.

d) $[C^+_{(aq)}]$ will decrease during the electrolytic process.

Section B: Short Answer Questions

ANSWERS: Rates and Equilibrium PAST SAC Questions

Question 1 Consider the equilibrium

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

Initially, 1.6 mol of SO_3 is placed in a sealed 3.0 L container. At equilibrium, $[O_2] = 0.15$ M

(a) find the concentration of the SO_3 at equilibrium

	$2SO_2$	$+ O_2$	$\rightleftharpoons 2SO_3$
n_i	0	0	1.6
n_f	.9	.45	1.6 − .9
conc.(eq)	.3	.15	.7/3

$[SO_3] = 0.23$ M (1)

(b) find the concentration of SO_2 at equilibrium

$[SO_2] = .30$ M (1)

(c) Use the above data to find the **value** of the equilibrium constant

$$K_{eq} = \frac{(.7/3)^2}{.3^2 \times .15}$$

$$= 4.0 \quad (1)$$

[3 marks]

Question 2 Two students were investigating the rate of the reaction between copper(II) carbonate and hydrochloric acid at a temperature of 4°C. They worked in a constant temperature room set at 4°C and added 0.500 mol of solid crushed blue-green copper(II) carbonate, $CuCO_3$ to 1.50 L of 0.100 M solution of hydrochloric acid, HCl in an open flask.

$$2HCl(aq) + CuCO_3(s) \rightarrow H_2O(l) + CO_2(g) + Cu^{2+}(aq) + 2Cl^-(aq)$$

(a) Explain why it was necessary to use an open flask

$CO_2(g)$ is a product, it must be released so only forward reaction occurs. (1) [1 mark]

(b) Explain how and why the changing concentration of the hydrochloric acid and the particle size of the crushed solid could affect the rate of the reaction.

higher conc. of HCl means more collisions (1)
smaller particle size " " " (1)
more collisions (if they are sufficiently energetic) means faster rate (1) [3 marks]

(c) The next day they repeated the experiment in the school laboratory using the same crushed copper(II) carbonate and the same acid. They found that the rate had increased.

Give a possible explanation for this discrepancy in terms of collision theory

- Room was warmer (1)
- at higher temperature particles move faster (1)
- more frequent and more energetic collisions (1) [3 marks]

Question 3

The most common method for the industrial production of hydrogen is the steam reforming process, which requires high temperature, high pressure and a Ni catalyst.
The equation for the reaction is

$$CH_4(g) + H_2O\,(g) \rightleftharpoons CO(g) + 3H_2(g) \quad \Delta H = +207\ KJ\ mol^{-1}$$

(a) Write an equilibrium expression for the steam reforming reaction.

$$K = \frac{[CO][H_2]^3}{[CH_4][H_2O]}$$ (1)

(b) Le Chatelier's principle indicates that the equilibrium yield for the reaction above is favoured by low pressure. Suggest one reason why high pressure is used in the industrial process described above.

higher pressure gives more favourable (faster) rate (1)

(c) At 1500°C the concentration of the gases in a particular equilibrium mixture were found to be $[CH_4] = 0.400$ M, $[CO] = 0.300$ M, $[H_2O] = 0.0680$ M
$K = 5.67\ M^2$ at 1500°C for the reaction

Calculate the molar concentration of H_2 in the equilibrium mixture.

$$\frac{.3 \times x^3}{.4 \times .068} = 5.67$$ (1)

$x = 0.801$

$[H_2] = 0.801$ M (1) (3 sig figures required)

[1 + 1 + 2 = 4 marks]

Question 4 Consider the equilibrium

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH\ (g) \quad \Delta H = -92\ KJ\ mol^{-1}$$

Complete the table below after the following changes are made and equilibrium is re-established.

Change to the system	Net Shift	Effect on the number of moles of H_2	Effect on the concentration of H_2
Addition of Helium	none	none	none
Decrease of pressure	back	↑	↓
Increase of temperature	back	↑	↑
Volume halved	forward	↓	↑
Methanol removed	forward	↓	↓

[5 marks]

5x (1) (for each group of any 3 correct responses

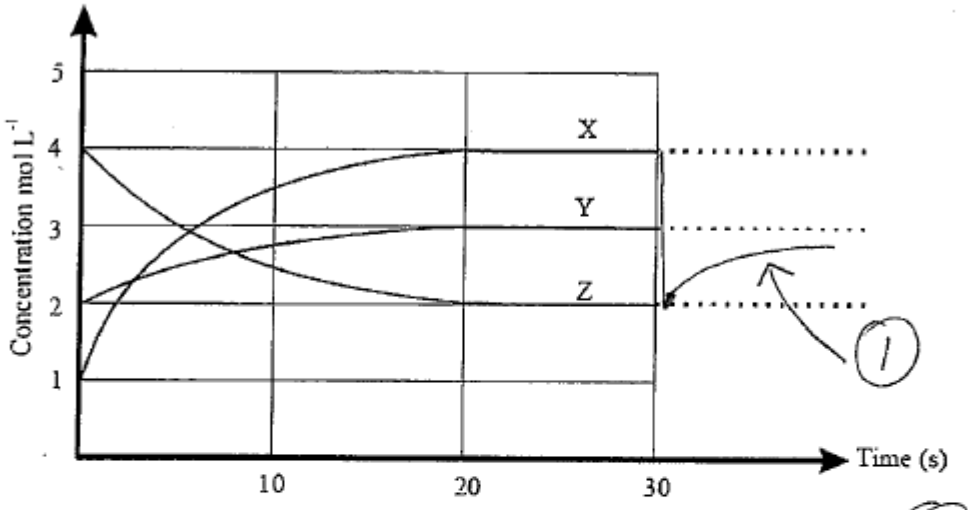
e.g 10 correct in total ⇒ 3 marks

" " " " "

8 " " 2 "

Question 5

The graph below shows the concentration, as a function of time, for a mixture of gases, X, Y and Z, added to a one-litre container and allowed to reach equilibrium



(a) How long does the mixture take to reach equilibrium? 20 s (1)

(b) (i) use the information provided on the graph to complete the table below

	Initial concentration	Equilibrium concentration	How the concentration changes as the system moves to equilibrium
X	1 M	4 M	↑ - gain 3 mol (1)
Y	2 M	3 M	↑ - gain 1 mol (1)
Z	4 M	2 M	↓ lose 2 mol (1)

(ii) Use the information in the table in (i) to deduce the equation for the equilibrium reaction

$$2\,Z_{(g)} \rightleftharpoons 3\,X_{(g)} + Y_{(g)}$$ (1)

(c) Write the equilibrium law expression and calculate the value of the equilibrium constant.

$$K_{eq} = \frac{[Y][X]^3}{[Z]^2} = 48$$ (1)

(d) At 30 seconds, the **volume of the container was doubled.** On the concentration time graph, sketch how the concentration of X would change immediately after and during establishment of a new equilibrium due to this volume increase.

[1 + 4 +1 + 1 = 7marks]

ANSWERS: Electrolysis PAST SAC questions

Question 6

a) Predict and write the reactions occurring at:

Cell 1

Electrode A: **Anode (+)** $2H_2O \longrightarrow O_2 + 4H^+ + 4e$

Electrode B: **Cathode (-)** $Fe^{3+} + e \longrightarrow Fe^{2+}$

Cell 2:

Electrode C: **Anode (+)** $2Br^- \longrightarrow Br_2 + 2e$

Electrode D: **Cathode (-)** $Zn^{2+} + 2e \longrightarrow Zn$

4 marks

b) When a current of 2.50 A was applied to the cell, 1.02 g of Zn was produced in Cell 2.

(i) Determine the amount, in mole, of electrons that passed through the cell.

$Zn^{2+} + 2e \longrightarrow Zn$

$n(Zn) = \frac{1.02}{65.4} = 0.0156\ mol$

n(e) = 2 x n(Zn) = 2 x 0.0156 mol = 0.0312 mol [1 mark]

(ii) Calculate the ***remaining*** concentration of Fe^{3+} (aq) in Cell 1 at this time.

$Fe^{3+} + e \longrightarrow Fe^{2+}$

$n(Fe^{3+})$reacted = n(e) = 0.0312 mole [1 mark]

$n(Fe^{3+})$left over = $n(Fe^{3+})$ initial - $n(Fe^{3+})$reacted [1 mark]
= (1 x 0.05) – 0.0312
= 0.0188 mol

$C(Fe^{3+})$ left over = $\frac{n}{v} = \frac{0.0188}{0.05} = 0.376\ M$ [1 mark]

(iii) How long (in minutes) would it have taken for the scenario in part (ii) to occur?

Q = n(e) X F = 0.0312 x 96500 = 3010 C [1 mark]

$t = \frac{Q}{I} = \frac{3010}{2.50} = 1204\ s = 20.1\ minutes$ [1 mark]

Total 10 marks